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Whispers in the air: Designing acoustic classifiers to detect fruit flies from afar

Alia Khalid [a], Muhammad Latif Anjum [a,*], Salman Naveed [b], Wajahat Hussain [a]

[a] National University of Sciences and Technology (NUST), Islamabad, Pakistan
[b] Department of Plant Breeding & Genetics, University of Agriculture, (UAF), Faisalabad, Pakistan

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ABSTRACT

Detecting weak wingbeats of a flying bug is a challenging problem in uncontrolled outdoor settings. In this work, we show that proper treatment of environmental noise is a key factor in robust acoustic classifier design and propose a novel environmental noise treatment method. Our proposed method generalizes over different classifiers and features. Our algorithm provides robust detection and classification of multiple bugs, over longest ranges reported, using simple microphones. In order to benchmark research in this area, we release a novel dataset containing acoustic data of four bugs (Guava fly, Melon fly, Blue bottle fly, and mosquitoes). We additionally investigate the feasibility of deploying our acoustic classifier on a noisy mobile platform, i.e., a drone. To this end, we expose the limitations of signal processing techniques to deal with loud drone noise. We demonstrate how soundproofing can be used to design acoustic sensing for drones.

1. Introduction

Can a farmer detect and classify a flying bug in the wild using a simple microphone? The wingbeats of flying bugs produce feeble sounds. Is it possible to detect these weak signals, in a noisy environment, at a reasonable distance from the bug? Such a direct bug monitoring has diverse applications [1–4].

It can help in timely deployment of a control strategy against various bugs. Firstly, over half a million people are killed by mosquito bites every year [5]. Few of the mosquito borne diseases have no cure [5]. Secondly, fruit flies are known for causing complete destruction to the fruits, leading to entire crop loss [6]. Using pesticides against fruit flies has harmful effects on the environment. Farmers in developing world rely mostly on pesticides to control pests, especially fruit flies. The use and misuse of these chemicals pose serious health and environmental risks [7–9]. Nearly three million cases of pesticides poisoning and 220,000 fatalities occur world-wide annually [10]. Due to these destructive environmental impacts of pesticides and increasing demand of hygienic food, new environmental friendly techniques to control fruit flies are arising [11]. Direct sensing of pests without trapping them has the advantage of being able to plan suitable intervention to repel them (and not killing them) away from the crop. This is of paramount importance considering the alarming 75% loss in insect biomass in the last three decades [12]. Sustainable cultivation always prefers pesticides-free control strategies [13–15].

Fly traps are the most common methods used for monitoring bugs [16]. There are traditional fly traps that attract bugs using a chemical [17,18]. Recently, smarter versions of these traps have been proposed. These smart traps raise alarms by automatic detection method as compared to manual inspection of the traditional fly traps [19]. However, these traps have limited surveillance area. In this work, our focus is to develop a sensing mechanism that works beyond the small trap and provides surveillance over a wider area, making the solution more effective.

Recently, a more flexible approach has shown that pointing the microphone of a simple cell phone towards the flying mosquitoes within 50 mm results in reliable detection and classification of multiple mosquito species [5]. The ability of simple microphones to detect bugs at a workable distance, greater than the aforementioned traps, along with their low cost makes them ideal for ubiquitous surveillance. In this work, we report reliable classification (>80%) of bugs at a distance of 300 mm (∼ 1 foot). To the best of our knowledge, this is the largest reported distance for flying bug classification in uncontrolled settings using low cost sensors. Our novel method for training acoustic classifiers is the reason for this improvement.

We are not the first ones to train robust acoustic classifiers. There is a large body of work on acoustic classifiers for detection and classification

* Corresponding author.
E-mail address: latif.anjum@seecs.edu.pk (M.L. Anjum).

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of various species. There have been two major areas of focus. The first area focuses on the best features for acoustic classification (FFT [20], MFCC [21], histogram of gradients (HoG) of FFTs [22] etc.). The second focused is finding the best classifiers for acoustic classification (neural networks for bats [23] and grass hoppers [24], Gaussian mixture models for crickets [25], regression trees for dolphins [26], deep learning for birds [27] and mosquitoes [28]). In this work, we demonstrate, for the first time, that another major factor that helps in reliable acoustic detection is the ability to handle the environment noise.

As we show in results, proper treatment of environment noise improves acoustic detection across multiple features and classifiers. Firstly, the weak acoustic signal (insect wing beats) must be collected in the presence of environmental noise. Secondly, the acquired signal must be preprocessed to remove the effect of the noise. These two steps make the classifier environment noise conscious. Note that the pre-processed signal is not totally noise free. Therefore, this makes this data acquisition strategy different to data captured in a noise-less environment.

Majority of the prior work propose a pre-processing stage of noise removal from the raw acoustic signal [5,23,26,24]. Few works also suggest to include noise in the training phase to improve classification robustness at test time [27]. There is no consensus on how to treat the environment noise. In this work, we provide detailed analysis of environment noise while designing acoustic classifiers.

To further increase the surveillance area, the microphone can be placed on a mobile platform (e.g., tractor or drone). How does the noise of the mobile platform affect the acoustic classifiers? Recently, drones have started to become an integral part of precision agriculture [29]. However, there is not much work done to design acoustic sensing mechanisms for these drones. The major reason is that the commercial low cost drones produce a considerable amount of noise.

One may assume that since the drone noise is stationary, and can be accurately modelled offline, it might be trivial to cancel this noise using adaptive filtering techniques [30]. In this work, we demonstrate that noise cancellation techniques fail to produce reliable classification results for the low SNR regime, i.e., weak fly wingbeats versus extremely loud drone noise. However, we show that muffling the drone noise using sound absorption material improves the acoustic classification accuracy.

2. Contributions

1. Robust algorithm for training acoustics classifier in noisy conditions.
2. Acoustic dataset of four types of bugs (fruit flies (Guava fly, Melon fly), Blue bottle fly and mosquitoes). To the best of our knowledge, this is the first public dataset for the mentioned fruit flies.[1]
3. Designing acoustic classifiers in the presence of noisy drone.

3. Related work

Detection of crop bugs and fruit flies using image has been well researched [31,32]. Marinho et al. [33] used X-ray images with convolutional neural networks to identify various species of pupal images. Similarly machine vision methods have been proposed for pest detection and classification at various stages growth in plant pathogens and Mediterranean fruit flies [34]. Zhou et al. presented a computer vision based method to study motion and defence mechanisms in fruit flies [35].

Imaging based methods depend on light, deteriorate under occlusion (bugs hiding behind the lead), and generally cameras have limited field of view. Acoustic signals are not affected by availability of light, show graceful degradation under occlusion and have 360° sensing capability. Furthermore, 1D audio signal processing is much more economical than 2D image processing.

[1] https://tinyurl.com/bugs-dataset.

Therefore, we leverage acoustic sensing to detect fruit flies and other bugs. Mankin et al. [36] used acoustic sensing to detect fruit flies in smart traps. They showed that the detector, designed in controlled anechoic chamber, was greatly affected in noisy field conditions. Acoustic sensing, for bug detection, fell out of favour due to inability to handle noise. Ali et al. presented an IoT enabled farm, aimed at capturing acoustic signals produced by fruit flies in order to detect presence of bugs [37]. A similar system using deep networks is presented in [38]. In this work, we propose novel treatment of environment noise while training acoustic classifiers for low SNR signals (insects wingbeats). Instead of removing the noise, our work suggests that acquiring the weak acoustic signal in a noisy environment improves the robustness of the classifier.

Photonic fences replaced acoustic sensors for smart traps. A laser projects light at the photosensitive screen [21,39,40]. A bug trying to cross the photonic fence produces shadow on the screen leading to its detection. These fences are highly accurate but they work in relatively controlled setups which limits their surveillance area. Our goal is to increase the surveillance area by dealing with the acoustic noise.

Beyond traps, Mukundarajan et al. [5] managed to detect mosquitoes in the open. Interestingly, they used cell phone microphones to detect mosquitoes within 50 mm (0.16 ft). They performed classification using Bhattacharyya distance between STFT of different mosquitoes. Unlike this approach, our goal is to monitor the effect of environmental noise on acoustic classifiers. We show that proper treatment of this noise, while designing acoustic classifiers, improves detection range.

Recently, deep learning has been used to detect mosquitoes using wingbeats [28]. However, training and testing was performed in controlled conditions. It is not clear how environment noise effects deep classifiers. We provide detailed analysis of environmental noise for designing acoustic classifiers. The need for direct sensing of bugs (mosquitoes) has even forced the development of deep visual classifiers [1]. It remains to be seen how these visual classifiers will work at night and under occlusion. Both these factors (occlusion, darkness) do not affect acoustic classification.

4. Classification in low SNR regime

The large environmental noise in practical applications such as detection of feeble acoustic signals of fruit flies *in the wild* significantly reduces SNR and consequently classification accuracy. We use four well-known classification algorithms (SVM, Decision Trees, k-Nearest neighbours, and Radial Basis Function) in combination with various features (FFT, HoG of FFT, MFCC) for designing our classification method. Different features and classifiers were selected to analyse the generalization ability of our approach. In order to handle large environmental noise resulting in low SNR signal, we use various data collection and noise removal strategies. Extensive experiments suggest that raw acoustic signal should be recorded in the field instead of controlled lab conditions. Classifiers trained on these data are more robust to environmental noise. Furthermore, removing environment noise at the pre-processing stage further improves the classification. Why record data in the noisy environment when noise will be removed later on? Noise removal only improves the SNR of the signal. It does not completely eradicate noise. Experiments show that the remaining noisy signal, with improved SNR, provides better classification results.

We carried out our analysis on four different types of bugs i.e., guava fly (B. Correcta [41]), melon fly (Bactrocera Cucurbitae [42]), blue bottle fly (Calliphora Vomitoria), and mosquitoes. We selected flying insects from two different applications, i.e., pest control (guava fly, melon fly) and disease control (mosquito). Blue bottle fly was selected as it is commonly available and its wing-beats may be confused with the other bugs of interest (guava fly, melon fly, mosquito). In this regard, we have four classes in our dataset. We test the following cases, and present their results in Section 5.

1. **Case 1: Train and Test in Controlled Environment:** In order to establish the benchmark for acoustic classification, we gathered the training and the test data in controlled indoor environment having minimal background noise. We gathered 400 data samples (100 samples for each class), which were split into training (70 samples for each class) and testing (30 samples for each class). Each training sample contained acoustic signals of only one bug, and was accordingly assigned class label.
2. **Case 2: Train in Controlled Environment and Test in the Wild:** We reused the training data from case 1 for the four bug classes. We added additional fifth class containing the background noise, so the data containing only noise can be classified into this category. Noise was modelled by recording audio signal in outdoor settings, in the absence of targeted four bugs. This background noise contained silence intervals interrupted by sounds of birds, humans and cars, etc., during the day and various species (frogs, crickets) during the night. Test data of the targeted four bugs was recorded in similar uncontrolled outdoor settings. Collection process of uncontrolled test data is provided in Section 4.1.
3. **Case 3: Train in Controlled Environment and Test in the Wild with Noise Removed:** Data collection in a controlled environment is logistically practical. Therefore, we would like to make the classifier, trained with these data, as robust as possible. For this case, we used the training data from the previous two cases. However, at test time we removed the noise. This noise was modelled by collecting data from test environment at a different day, time and location. We used two softwares, i.e., Audacity® and Adobe Audition® for noise removal. Audacity® works by asking the users to provide noise profile by indicating only noise. After that, the noise profile is used to remove the noise from the entire audio file. Adobe Audition® provides similar tools. We used adaptive noise reduction available in Adobe Audition®.
4. **Case 4: Train and Test in the Wild:** For this case, we gathered the train and test data in the uncontrolled outdoor environment. No noise was removed during training and test time. We wanted to test how good raw noisy signal is for classification.
5. **Case 5: Train and Test in the Wild with Noise Removed:** We used the data from case 4. However, we removed noise both at training and test time. As experiments indicate this is the best practice for training acoustic classifiers in noisy environments.
6. **Case 6: Train and Test in the Wild in the Presence of Drone:** In order to check whether this acoustic setup can be mounted on a mobile platform, we checked the performance of our classifier, from case 5, in the presence of drone.

4.1. Data collection

We acquired the fruit flies (guava fly, melon fly) from two labs having fruit fly rearing chambers. Blue bottle flies and mosquitoes were captured from the park. In order to capture the data in controlled settings, the bugs were placed in a container along with the recording device. The bugs died after a short while. For capturing the data in noisy settings, we replayed their recorded wingbeats outdoors. Recording these replayed wingbeats added background noise in the audio signals. We ensured that the amplitude of the replayed signal was not louder than the original recordings in controlled settings (compare the frequency histogram given in the first two rows of Fig. 1). Dataset details are given in Table 1. Same amount of data was collected for night time experiments as well.

4.2. Experimental setup

We used Respeaker® 4x microphone array (Fig. 5a) for audio recording. Only single microphone was used for classification experiments. We used Focus drone. Plastic and foam were used as sound proofing material in drone experiments. Nokia Asha 308 mobile was used for replaying the recorded wingbeat sounds. Unless stated otherwise the distance between sound source and the microphone is 1 foot.

Table 1
Acoustic Dataset Details. Each training/test sample is 1 second long. Equal amount of training/test data was captured for night time experiments as well for cases 2-6.

Cases	Classes	Train Set Size	Test Set Size
1	4 bugs	280(70/class)	120(30/class)
2	4 bugs + Background noise	500(100/class)	450(90/class)
3	4 bugs + Background noise	500(100/class)	450(90/class)
4	4 bugs + Background noise	1000(200/class)	450(90/class)
5	4 bugs + Background noise	1000(200/class)	450(90/class)
6	4 bugs + Background noise	1000(200/class)	450(90/class)

5. Experiments

5.1. How to train acoustic classifiers in noisy conditions?

The classification results for the first five cases are given as confusion matrices in Fig. 2. For this experiment we used FFT as feature and SVM as classifier. In order to analyse the effect of different strategies for acoustic classifier training, we fixed the feature (FFT) and classifier (SVM).

Classifier tested at day time was trained using data from day time. Same goes for night time classification. Columns indicate the input class and rows show the classifier output class. Darker shades represent higher number of instances in a given matrix cell. Ideally, the dark shades should be along the diagonal as in case 1 in Fig. 2. Case 6 (involving drone) will be discussed separately.

The classifier from case 1, when tested in noisy conditions, confuses the majority of the test examples as noise (case 2 in Fig. 2). Majority of the test examples are labelled as background noise. Removing the environmental noise at test time improves the classification as dark shades return to diagonal (case 3 in Fig. 2). Training and testing with raw noisy signal, without noise removal, again confuses the classifier (case 4 in Fig. 2). The best results are achieved when training and testing are performed in outdoor conditions with noise removal (compare case 1 and case 5 in Fig. 2).

5.2. Generalization over different features and classifiers?

Fig. 3 shows that our proposed strategy of training and testing in noisy conditions with noise removal (case 5) performs best for **different features** (FFT, HoG of FFT, MFCC), **different classifiers** (SVM, KNN, Radial Basis Function (RBF) and Decision Trees (DT)) and **different times** of the day.

Each bar in Fig. 3 represents the classification accuracy of one of the five cases discussed. The fifth bar, in each sub figure, is closest to the first bar. The second best performing scenario is case 3 (third bar in Fig. 3). These results can be corroborated from Fig. 2 as well. SVM, in combination with FFT, performs the best both during day (84.86%) and night time (92.4%). The second best performing combo is DT with FFT, i.e., 82.42% (day) and 88.92% (night).

5.3. Best features and classifiers for long range detection?

In order to test the range of our direct sensing, we performed experiments for case 5, with multiple distances (1-3 foot) between the wingbeat source and the microphone. Firstly, we fixed the best performing feature, i.e., FFT, and evaluated the performance of different classifiers. Fig. 4 shows that all the classifiers show similar degradation in performance.

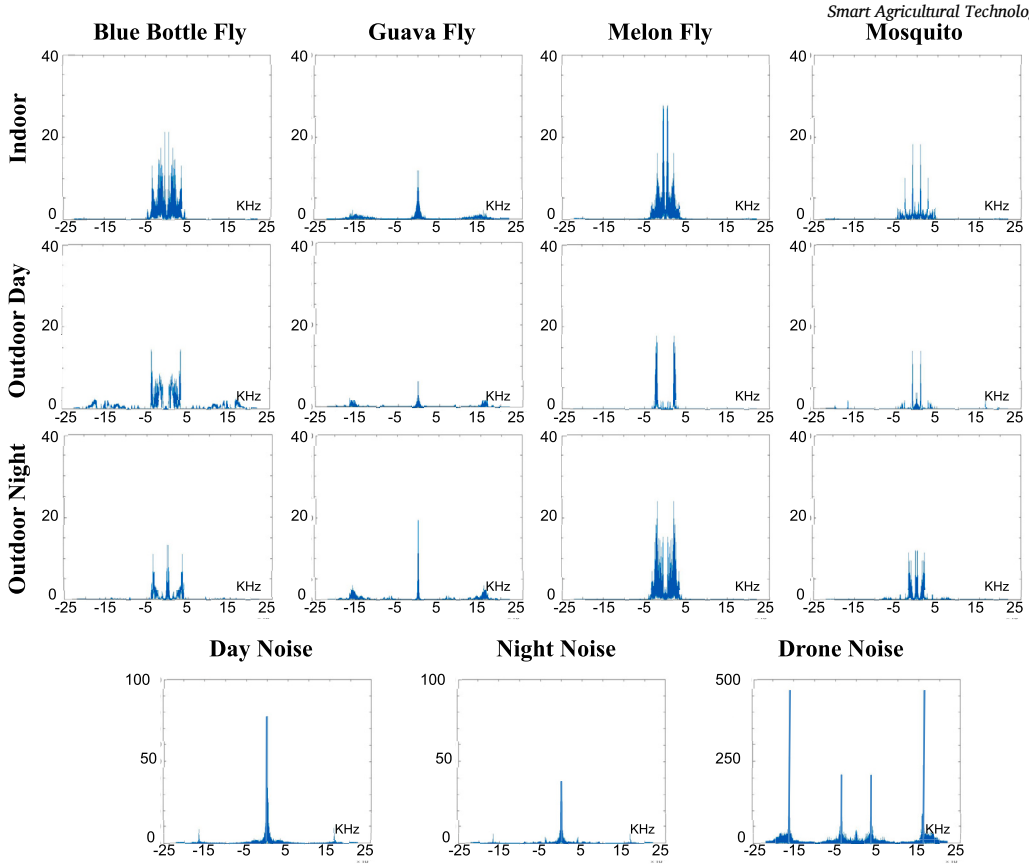

Fig. 1. Comparison of frequency histograms of bug wingbeats, background noise and the drone noise. Overlapping frequency spectrums of four bugs, background noise and the drone noise. Environment noise is louder (2x for night, 8x for day) and drone noise is 20x louder than wingbeats.

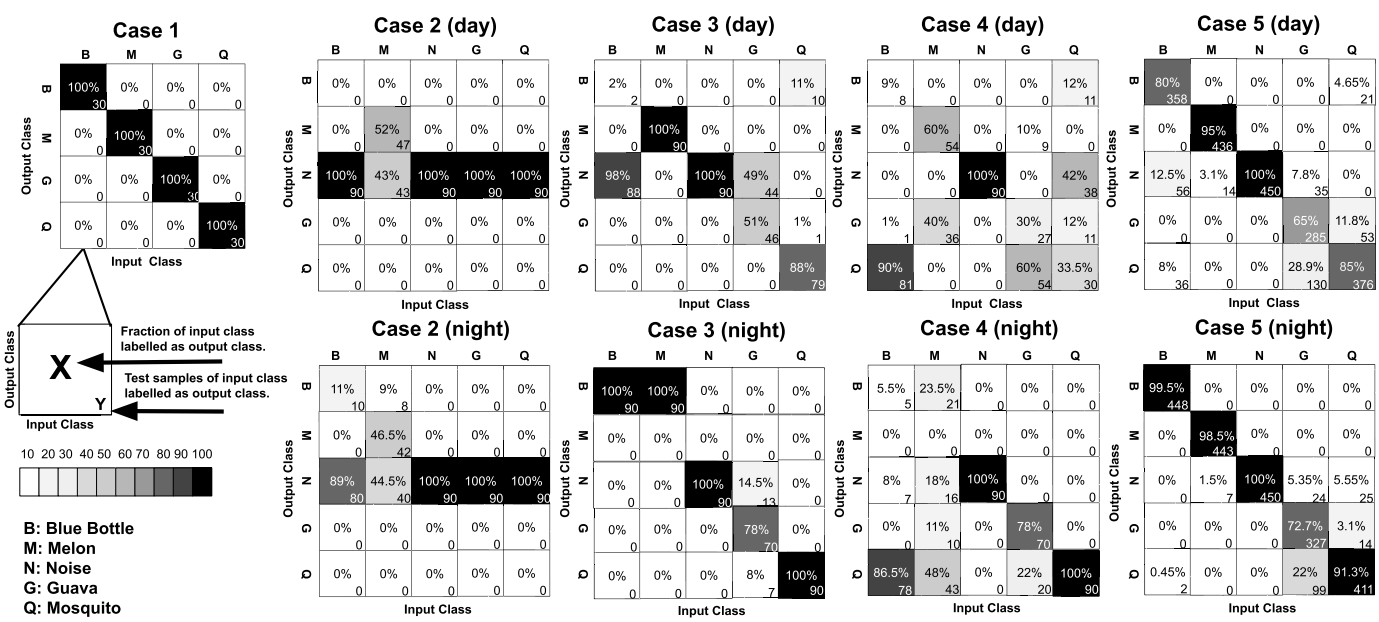

Fig. 2. Novel treatment of environment noise: Confusion matrices indicate that our proposed algorithm (case 5) is closest to ideal performance (case 1). Each entry of confusion matrix indicates fraction of input class (target class) being labelled as output class. Darker colour means greater fraction of the input class being labelled as a specific output class. Ideally, darker colour should be along the diagonal.

To our surprise, the accuracy is higher at night time (~90% at 1 ft range) as compared to day time (~82%). The common experience is that during night multiple bugs produce noise. Analysing the audio signal revealed that, at night time there may be multiple noise sources (crickets, frogs) but they are stationary (repeat every day). Hence, noise removal step takes care of multiple sources. Whereas, during the day, the noise sources are non-stationary (different car horns, different birds).

Furthermore, we fixed the classifier (SVM) and tested different features. FFTs perform better at smaller ranges whereas MFCCs perform better at long ranges.

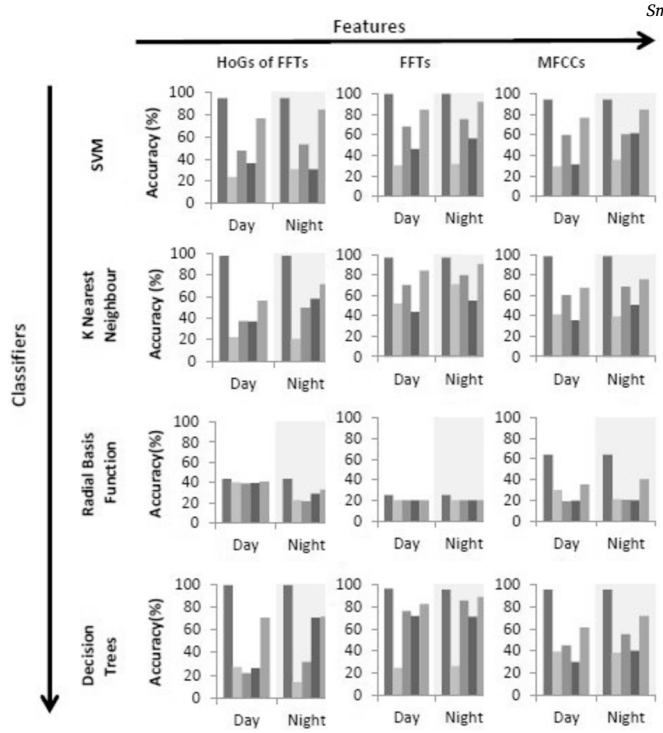

Fig. 3. Our novel algorithm generalizes for different features, classifiers and day times. Proposed algorithm (5st bar) is closest to ideal performance (1st bar). The 1st bar corresponds to case 1, 2nd to case 2, ..., 5th to case 5, for both day and night.

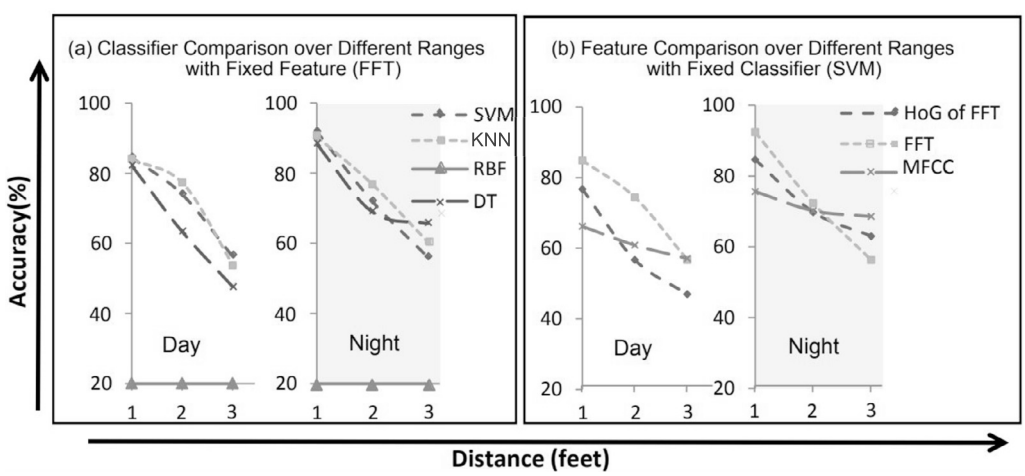

Fig. 4. Our proposed method produces robust detection over longer ranges. ~90% at night and ~82% during day at 1 ft range. This is largest range reported so far for low cost microphones.

5.4. How to use drone for acoustic applications?

We tested the best performing combination (SVM + FFT + case 5 classifier) in the presence of drone. Due to high drone noise, classification accuracy was very low. Therefore, in addition to environment noise, we tried to remove the drone noise at test time. This resulted in flat audio signal (Fig. 5b). Adaptive noise removal failed due to high energy of drone noise as compared to feeble wingbeat signal. The accuracy dipped to 20% (second bar in Fig. 5d) since everything was labelled as silence (the fifth class) for these flat test signals.

After failure of signal processing methods to remove drone noise, we tried to change the physical setup (Fig. 5a). Increasing the distance between the drone and the microphone improved the results slightly (third bar in Fig. 5d). Next using the sound proof material (foam, plastic) between drone and the microphone (Fig. 5) further increased the performance (fourth bar in Fig. 5d).

The best results were achieved when training and testing were performed, in the presence of drone, with hurdles between the microphone and the drone (the fifth bar in Fig. 5d). Drone noise was removed both at test and train time. This is consistent with our proposed strategy of proper treatment of environmental noise.

6. Conclusion

We have proposed a novel method to train acoustic detectors in low SNR regime. Environmental noise helps acoustic classifiers if treated properly. Our proposed method produces robust wing-beat classification over longer ranges, i.e., ~90% at night and ~82% during day at 1 ft range. This is largest range reported so far for low cost microphones. SVM in combination with FFT performs the best for our proposed strategy.

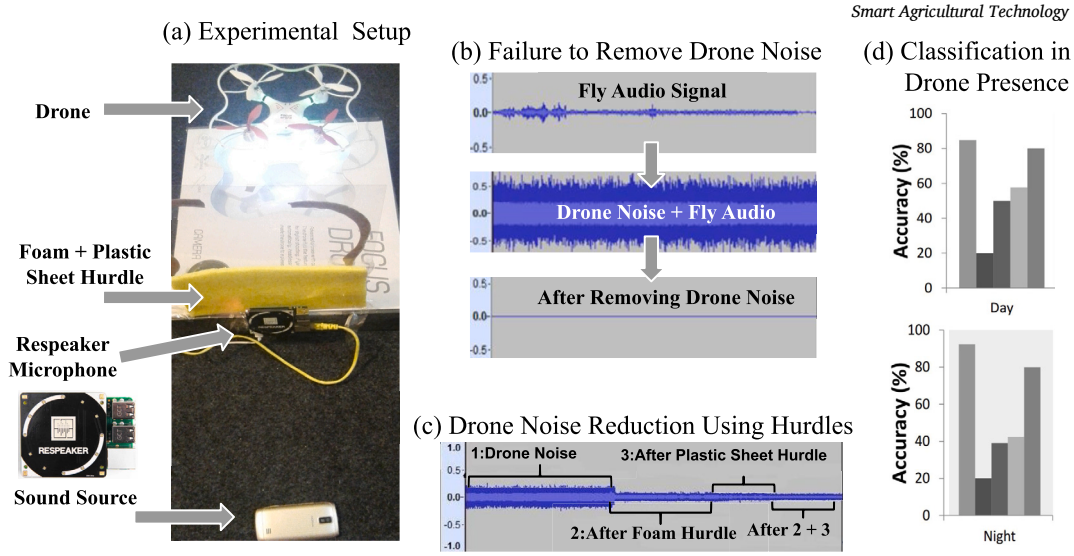

Fig. 5. How to design acoustic application for drones? (a) Experimental setup including wingbeat source (mobile phone), Respeaker microphone, drone and the soundproofing hurdles. (b) Removing drone noise from raw wingbeat signal, without using hurdles, leaves no information (flat audio signal). (c) Placing hurdles between drone and the microphone reduces the drone noise. (d) Detecting bugs in the presence of drone. Our proposed approach (5th bar) is closest to best achievable case 5 (1st bar). For detail check text.

This finding is generic and can be applied in other low SNR scenarios, e.g., natural sound recordings made in outdoor environments, such as in forests [43] or near bodies of water, where wind, wildlife, or other environmental sounds dominate the desired audio.

In order to develop acoustic applications for commercial noisy drones, better ways to muffle drone noise are required, e.g., low noise drone rotor design [44]. To best of our knowledge, this is the first work exposing this issue in the context of designing acoustic classifiers for commercial drones.

CRediT authorship contribution statement

Alia Khalid: Software, Investigation, Data curation. **Muhammad Latif Anjum:** Writing – review & editing, Writing – original draft, Supervision. **Salman Naveed:** Visualization, Supervision. **Wajahat Hussain:** Writing – original draft, Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this article.

Data availability

Dataset is uploaded, and link is provided within the manuscript.

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